Chapter 2

Light and Shadows

We can see things only when there is light. When light from an object reaches our eyes, we see the object. In a dark room, we cannot see anything at all.

Sources of light

Things that give out their own light are called sources of light. The Sun, a burning candle, an electric bulb and a firefly are sources of light. The Sun is our biggest source of light.

The Moon is not a source of light. It looks bright at night because sunlight falls on it and bounces back towards the Earth.

Light travels in a straight line

Take three cards and make a small hole in the middle of each. Stand them in a row and light a candle at one end. Look through the holes from the other end. You can see the flame only when all the holes are in a straight line. If you move one card a little, the flame disappears. This shows that light travels in a straight line.

Can light pass through everything?

- Transparent things let almost all light pass through. Example: clear glass, clean water.
- Translucent things let only some light pass through. Example: butter paper, frosted glass.
- Opaque things do not let any light pass through. Example: wood, a book, our body.

Shadows

When an opaque object blocks light, a dark area forms behind it. This dark area is called a shadow. Because light travels in a straight line, it cannot bend around the object, and so a shadow is formed.

- A shadow always forms on the side opposite to the source of light.
- A shadow has the shape of the outline of the object, but it shows no colours.
- In the morning and evening, when the Sun is low in the sky, our shadows are long. At noon, when the Sun is high overhead, our shadows are very short.

Activity: Shadow puppets

Shine a torch on a wall in a dark room. Hold your hands between the torch and the wall and make shapes like a bird or a dog. Move your hands closer to the torch and the shadow becomes bigger.

Mirrors

A mirror is very smooth and shiny. When light falls on a mirror, it bounces back. This bouncing back of light is called reflection. Because of reflection, we can see our image in a mirror.

Let's think

1. Is the Moon a source of light? Why?
2. Why are our shadows short at noon?
3. Give one example each of a transparent, translucent and opaque object.

New words

opaque - not letting any light pass through
reflection - the bouncing back of light from a surface